

# Safety data sheet Safety Data Sheet

acc. to Regulation (EC) No. 1907/2006 (REACH)



## 4-Chloroaniline ≥98 %, for synthesis

article number: **1C3A**

Version: **4.0 en**

Replaces version of: 2024-09-18

Version: (3)

date of compilation: 2020-05-26

Revision: 2025-08-04

## SECTION 1: Identification of the substance/mixture and of the company/undertaking

### 1.1 Product identifier

|                                 |                                             |
|---------------------------------|---------------------------------------------|
| Identification of the substance | <b>4-Chloroaniline</b> ≥98 %, for synthesis |
| Article number                  | 1C3A                                        |
| Index No (GB CLP)               | 612-137-00-9                                |
| EC number                       | 203-401-0                                   |
| CAS number                      | 106-47-8                                    |

### 1.2 Relevant identified uses of the substance or mixture and uses advised against

|                           |                                                                                                                                                      |
|---------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|
| Relevant identified uses: | Laboratory and analytical use<br>Laboratory chemical                                                                                                 |
| Uses advised against:     | Do not use for products which come into contact with foodstuffs. Do not use for private purposes (household). Food, drink and animal feeding-stuffs. |

### 1.3 Details of the supplier of the safety data sheet

Carl Roth GmbH + Co. KG  
Schoemperlenstr. 3-5  
D-76185 Karlsruhe  
Germany

**Telephone:** +49 (0) 721 - 56 06 0

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**e-mail:** [sicherheit@carlroth.de](mailto:sicherheit@carlroth.de)

**Website:** [www.carlroth.de](http://www.carlroth.de)

Competent person responsible for the safety data sheet: Department Health, Safety and Environment

**e-mail (competent person):** [sicherheit@carlroth.de](mailto:sicherheit@carlroth.de)

### 1.4 Emergency telephone number

| Name                                                  | Street    | Postal code/city     | Telephone    | Website |
|-------------------------------------------------------|-----------|----------------------|--------------|---------|
| National Poisons Information Service<br>City Hospital | Dudley Rd | B187QH<br>Birmingham | 844 892 0111 |         |

## SECTION 2: Hazards identification

### 2.1 Classification of the substance or mixture

Classification acc. to GHS

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| Section | Hazard class                                          | Cat-egory | Hazard class and category | Hazard statement |
|---------|-------------------------------------------------------|-----------|---------------------------|------------------|
| 3.1O    | Acute toxicity (oral)                                 | 3         | Acute Tox. 3              | H301             |
| 3.1D    | Acute toxicity (dermal)                               | 3         | Acute Tox. 3              | H311             |
| 3.1I    | Acute toxicity (inhal.)                               | 3         | Acute Tox. 3              | H331             |
| 3.4S    | Skin sensitisation                                    | 1         | Skin Sens. 1              | H317             |
| 3.6     | Carcinogenicity                                       | 1B        | Carc. 1B                  | H350             |
| 4.1A    | Hazardous to the aquatic environment - acute hazard   | 1         | Aquatic Acute 1           | H400             |
| 4.1C    | Hazardous to the aquatic environment - chronic hazard | 1         | Aquatic Chronic 1         | H410             |

For full text of abbreviations: see SECTION 16

### The most important adverse physicochemical, human health and environmental effects

Spillage and fire water can cause pollution of watercourses.

## 2.2 Label elements

### Labelling

#### Signal word

**Danger**

#### Pictograms

GHS06, GHS08,  
GHS09



### Hazard statements

H301+H311+H331 Toxic if swallowed, in contact with skin or if inhaled  
H317 May cause an allergic skin reaction  
H350 May cause cancer  
H410 Very toxic to aquatic life with long lasting effects

### Precautionary statements

#### Precautionary statements - prevention

P273 Avoid release to the environment  
P280 Wear protective gloves/protective clothing/eye protection/face protection

#### Precautionary statements - response

P301+P310 IF SWALLOWED: Immediately call a POISON CENTER/doctor  
P302+P352 IF ON SKIN: Wash with plenty of soap and water  
P304+P340 IF INHALED: Remove person to fresh air and keep comfortable for breathing  
P308+P313 IF exposed or concerned: Get medical advice/attention

For professional users only

## 2.3 Other hazards

### Results of PBT and vPvB assessment

According to the results of its assessment, this substance is not a PBT or a vPvB.

### Endocrine disrupting properties

Does not contain an endocrine disruptor (ED) at a concentration of  $\geq 0,1\%$ .

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## SECTION 3: Composition/information on ingredients

### 3.1 Substances

|                   |                                   |
|-------------------|-----------------------------------|
| Name of substance | 4-Chloroaniline                   |
| Molecular formula | C <sub>6</sub> H <sub>6</sub> ClN |
| Molar mass        | 127,6 g/mol                       |
| CAS No            | 106-47-8                          |
| EC No             | 203-401-0                         |
| Index No (GB CLP) | 612-137-00-9                      |

| Substance, Specific Conc. Limits, M-factors, ATE |           |                                        |                                            |
|--------------------------------------------------|-----------|----------------------------------------|--------------------------------------------|
| Specific Conc. Limits                            | M-Factors | ATE                                    | Exposure route                             |
| -                                                | -         | 100 mg/kg<br>300 mg/kg<br>>0,5 mg/l/4h | oral<br>dermal<br>inhalation:<br>dust/mist |

## SECTION 4: First aid measures

### 4.1 Description of first aid measures



#### General notes

Take off immediately all contaminated clothing. Self-protection of the first aider.

#### Following inhalation

Call a physician immediately. If breathing is irregular or stopped, administer artificial respiration.

#### Following skin contact

Rinse skin with water/shower. After contact with skin, wash immediately with plenty of water. In case of skin reactions, consult a physician.

#### Following eye contact

Rinse cautiously with water for several minutes. In all cases of doubt, or when symptoms persist, seek medical advice.

#### Following ingestion

Rinse mouth immediately and drink plenty of water. In case of accident or unwellness, seek medical advice immediately (show directions for use or safety data sheet if possible).

### 4.2 Most important symptoms and effects, both acute and delayed

Has degreasing effect on the skin, Irritant effects, Allergic reactions, Nausea, Vomiting, Headache, Cough, Spasms, Dyspnoea, Blood pressure drop, Cardiac arrhythmias, Methaemoglobinaemia, Cyanosis (blue coloured blood)

### 4.3 Indication of any immediate medical attention and special treatment needed

none

## SECTION 5: Firefighting measures

### 5.1 Extinguishing media

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### Suitable extinguishing media

co-ordinate firefighting measures to the fire surroundings!  
water, foam, alcohol resistant foam, dry extinguishing powder, ABC-powder

### Unsuitable extinguishing media

water jet

## 5.2 Special hazards arising from the substance or mixture

Combustible. Deposited combustible dust has considerable explosion potential.

### Hazardous combustion products

In case of fire may be liberated: Nitrogen oxides (NO<sub>x</sub>), Carbon monoxide (CO), Carbon dioxide (CO<sub>2</sub>), Hydrogen chloride (HCl)

## 5.3 Advice for firefighters

In case of fire and/or explosion do not breathe fumes. Do not allow firefighting water to enter drains or water courses. Fight fire with normal precautions from a reasonable distance. Wear self-contained breathing apparatus. Wear full chemical protective clothing.

## SECTION 6: Accidental release measures

### 6.1 Personal precautions, protective equipment and emergency procedures



#### For non-emergency personnel

Use personal protective equipment as required. Avoid contact with skin, eyes and clothes. Do not breathe dust.

### 6.2 Environmental precautions

Keep away from drains, surface and ground water. Retain contaminated washing water and dispose of it. If substance has entered a water course or sewer, inform the responsible authority.

### 6.3 Methods and material for containment and cleaning up

#### Advice on how to contain a spill

Covering of drains. Take up mechanically.

#### Advice on how to clean up a spill

Take up mechanically. Control of dust.

#### Other information relating to spills and releases

Place in appropriate containers for disposal. Ventilate affected area.

### 6.4 Reference to other sections

Hazardous combustion products: see section 5. Personal protective equipment: see section 8. Incompatible materials: see section 10. Disposal considerations: see section 13.

## SECTION 7: Handling and storage

### 7.1 Precautions for safe handling

Provision of sufficient ventilation. Use extractor hood (laboratory). Handle and open container with care. Avoid exposure. Avoid dust formation. Clear contaminated areas thoroughly.

#### Measures to prevent fire as well as aerosol and dust generation

Removal of dust deposits.

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### Specific notes/details

Dust deposits may accumulate on all deposition surfaces in a technical room. The product in the delivered form is not dust explosion capable; the enrichment of fine dust however leads to the danger of dust explosion.

### Measures to protect the environment

Avoid release to the environment.

### Advice on general occupational hygiene

When using do not eat or drink. Thorough skin-cleansing after handling the product.

## 7.2 Conditions for safe storage, including any incompatibilities

Store in a dry place. Keep container tightly closed. May cause decomposition by long-term light influence.

### Incompatible substances or mixtures

Observe hints for combined storage. Incompatible materials: see section 10.

### Protect against external exposure, such as

direct light irradiation, UV-radiation/sunlight, contact with air/oxygen

### Consideration of other advice:

Store locked up.

### Ventilation requirements

Keep any substance that emits harmful vapours or gases in a place that allows these to be permanently extracted. Use local and general ventilation.

### Specific designs for storage rooms or vessels

Recommended storage temperature: 15 – 25 °C

## 7.3 Specific end use(s)

No information available.

## SECTION 8: Exposure controls/personal protection

### 8.1 Control parameters

#### National limit values

#### Occupational exposure limit values (Workplace Exposure Limits)

| Country | Name of agent | CAS No | Identifier | TWA<br>[mg/m <sup>3</sup> ] | STEL<br>[mg/m <sup>3</sup> ] | Ceiling-C<br>[mg/m <sup>3</sup> ] | Notation | Source    |
|---------|---------------|--------|------------|-----------------------------|------------------------------|-----------------------------------|----------|-----------|
| GB      | dust          |        | WEL        | 10                          |                              |                                   | i        | EH40/2005 |
| GB      | dust          |        | WEL        | 4                           |                              |                                   | r        | EH40/2005 |

#### Notation

|           |                                                                                                                                                                          |
|-----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Ceiling-C | Ceiling value is a limit value above which exposure should not occur                                                                                                     |
| i         | Inhalable fraction                                                                                                                                                       |
| r         | Respirable fraction                                                                                                                                                      |
| STEL      | Short-term exposure limit: a limit value above which exposure should not occur and which is related to a 15-minute period (unless otherwise specified)                   |
| TWA       | Time-weighted average (long-term exposure limit): measured or calculated in relation to a reference period of 8 hours time-weighted average (unless otherwise specified) |

### 8.2 Exposure controls

#### Individual protection measures (personal protective equipment)

#### Eye/face protection

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Use safety goggle with side protection.

### Skin protection



#### • hand protection

Wear suitable gloves. Chemical protection gloves are suitable, which are tested according to EN 374. Check leak-tightness/impermeability prior to use. For special purposes, it is recommended to check the resistance to chemicals of the protective gloves mentioned above together with the supplier of these gloves. The times are approximate values from measurements at 22 ° C and permanent contact. Increased temperatures due to heated substances, body heat etc. and a reduction of the effective layer thickness by stretching can lead to a considerable reduction of the breakthrough time. If in doubt, contact manufacturer. At an approx. 1.5 times larger / smaller layer thickness, the respective breakthrough time is doubled / halved. The data apply only to the pure substance. When transferred to substance mixtures, they may only be considered as a guide.

#### • type of material

Butyl caoutchouc (butyl rubber)

#### • material thickness

$\geq 0,5$  mm

#### • breakthrough times of the glove material

>480 minutes (permeation: level 6)

#### • other protection measures

Take recovery periods for skin regeneration. Preventive skin protection (barrier creams/ointments) is recommended.

### Respiratory protection



Respiratory protection necessary at: Dust formation. Particulate filter device (EN 143). P3 (filters at least 99,95 % of airborne particles, colour code: White). Type: A-P2 (combined filters against particles and organic gases and vapours, colour code: Brown/White).

### Environmental exposure controls

Keep away from drains, surface and ground water.

## SECTION 9: Physical and chemical properties

### 9.1 Information on basic physical and chemical properties

|                                                          |                                                   |
|----------------------------------------------------------|---------------------------------------------------|
| Physical state                                           | solid                                             |
| Colour                                                   | colourless - light yellow                         |
| Odour                                                    | characteristic                                    |
| Melting point/freezing point                             | $\sim 70$ °C                                      |
| Boiling point or initial boiling point and boiling range | 232 °C                                            |
| Flammability                                             | this material is combustible, but will not ignite |

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|                                                     |                                                                 |
|-----------------------------------------------------|-----------------------------------------------------------------|
|                                                     | readily                                                         |
| Lower and upper explosion limit                     | not relevant (solid)                                            |
| Flash point                                         | 123 °C (c.c.)                                                   |
| Auto-ignition temperature                           | not determined                                                  |
| Decomposition temperature                           | 330 °C                                                          |
| pH (value)                                          | not applicable                                                  |
| Kinematic viscosity                                 | not relevant                                                    |
| <u>Solubility(ies)</u>                              |                                                                 |
| Water solubility                                    | 3,9 g/l (ECHA)                                                  |
| <u>Partition coefficient</u>                        |                                                                 |
| Partition coefficient n-octanol/water (log value):  | this information is not available                               |
| Vapour pressure                                     | 0,07 hPa                                                        |
| <u>Density and/or relative density</u>              |                                                                 |
| Density                                             | 1,43 g/cm <sup>3</sup> at 20 °C 1,17 g/cm <sup>3</sup> at 70 °C |
| Relative vapour density                             | not relevant (solid)                                            |
| Bulk density                                        | ~800 kg/m <sup>3</sup>                                          |
| Particle characteristics                            | No data available.                                              |
| <u>Other safety parameters</u>                      |                                                                 |
| Oxidising properties                                | none                                                            |
| <b>9.2 Other information</b>                        |                                                                 |
| Information with regard to physical hazard classes: | hazard classes acc. to GHS (physical hazards): not relevant     |
| Other safety characteristics:                       | There is no additional information.                             |

## SECTION 10: Stability and reactivity

### 10.1 Reactivity

The product in the delivered form is not dust explosion capable; the enrichment of fine dust however leads to the danger of dust explosion.

### 10.2 Chemical stability

May cause decomposition by long-term light influence.

### 10.3 Possibility of hazardous reactions

**Violent reaction with:** strong oxidiser, Carboxylic acid anhydride, Acetic anhydride, Acids

### 10.4 Conditions to avoid

Direct light irradiation. UV-radiation/sunlight. Keep away from heat. Decomposition takes place from

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temperatures above: 330 °C.

### 10.5 Incompatible materials

There is no additional information.

### 10.6 Hazardous decomposition products

Hazardous combustion products: see section 5.

## SECTION 11: Toxicological information

### 11.1 Information on toxicological effects

#### Classification acc. to GHS

##### Acute toxicity

Toxic if swallowed. Toxic in contact with skin. Toxic if inhaled.

##### Skin corrosion/irritation

Shall not be classified as corrosive/irritant to skin.

##### Serious eye damage/eye irritation

Shall not be classified as seriously damaging to the eye or eye irritant.

##### Respiratory or skin sensitisation

May cause an allergic skin reaction.

##### Germ cell mutagenicity

Shall not be classified as germ cell mutagenic.

##### Carcinogenicity

May cause cancer.

##### Reproductive toxicity

Shall not be classified as a reproductive toxicant.

##### Specific target organ toxicity - single exposure

Shall not be classified as a specific target organ toxicant (single exposure).

##### Specific target organ toxicity - repeated exposure

Shall not be classified as a specific target organ toxicant (repeated exposure).

##### Aspiration hazard

Shall not be classified as presenting an aspiration hazard.

##### Endocrine disruptor for human health

Shall not be classified as an endocrine disruptor for human health.

#### Symptoms related to the physical, chemical and toxicological characteristics

##### • If swallowed

vomiting, nausea

##### • If in eyes

causes slight to moderate irritation

##### • If inhaled

cough, headache, Dyspnoea

##### • If on skin

Prolonged or repeated skin contact may cause removal of natural fat from the skin resulting in dermatitis (skin inflammation), May produce an allergic reaction, pruritis, localised redness

##### • Other information

Other adverse effects: Liver and kidney damage, Cardiac arrhythmias, Blood pressure drop, Spasms, Methaemoglobinaemia, Cyanosis (blue coloured blood)



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### 11.2 Information on other hazards

There is no additional information.

## SECTION 12: Ecological information

### 12.1 Toxicity

Very toxic to aquatic life with long lasting effects.

### 12.2 Persistence and degradability

Theoretical Oxygen Demand (without nitrification): 1,63 mg/mg

Theoretical Oxygen Demand (with nitrification): 2,132 mg/mg

Theoretical Carbon Dioxide: 2,07 mg/mg

### 12.3 Bioaccumulative potential

Data are not available.

### 12.4 Mobility in soil

Data are not available.

### 12.5 Results of PBT and vPvB assessment

According to the results of its assessment, this substance is not a PBT or a vPvB.

### 12.6 Endocrine disrupting properties

Does not contain an endocrine disruptor (ED) at a concentration of  $\geq 0,1\%$ .

### 12.7 Other adverse effects

Data are not available.

## SECTION 13: Disposal considerations

### 13.1 Waste treatment methods



This material and its container must be disposed of as hazardous waste. Dispose of contents/container in accordance with local/regional/national/international regulations.

#### Sewage disposal-relevant information

Do not empty into drains. Avoid release to the environment. Refer to special instructions/safety data sheets.

#### Waste treatment of containers/packagings

It is a dangerous waste; only packagings which are approved (e.g. acc. to ADR) may be used. Handle contaminated packages in the same way as the substance itself. Completely emptied packages can be recycled.

### 13.2 Relevant provisions relating to waste

The allocation of waste identity numbers/waste descriptions must be carried out according to the EEC, specific to the industry and process.

#### Properties of waste which render it hazardous

HP 6 acute toxicity

HP 7 carcinogenic

HP 13 sensitising

HP 14 ecotoxic

### 13.3 Remarks

Waste shall be separated into the categories that can be handled separately by the local or national waste management facilities. Please consider the relevant national or regional provisions. Non-contaminated packages may be recycled.

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## SECTION 14: Transport information

### 14.1 UN number or ID number

|           |         |
|-----------|---------|
| ADRRID    | UN 2018 |
| IMDG-Code | UN 2018 |
| ICAO-TI   | UN 2018 |

### 14.2 UN proper shipping name

|           |                       |
|-----------|-----------------------|
| ADRRID    | CHLOROANILINES, SOLID |
| IMDG-Code | CHLOROANILINES, SOLID |
| ICAO-TI   | Chloroanilines, solid |

### 14.3 Transport hazard class(es)

|           |     |
|-----------|-----|
| ADRRID    | 6.1 |
| IMDG-Code | 6.1 |
| ICAO-TI   | 6.1 |

### 14.4 Packing group

|           |    |
|-----------|----|
| ADRRID    | II |
| IMDG-Code | II |
| ICAO-TI   | II |

### 14.5 Environmental hazards

hazardous to the aquatic environment

### 14.6 Special precautions for user

Provisions for dangerous goods (ADR) should be complied within the premises.

### 14.7 Maritime transport in bulk according to IMO instruments

The cargo is not intended to be carried in bulk.

### 14.8 Information for each of the UN Model Regulations

#### Agreement concerning the International Carriage of Dangerous Goods by Road (ADR) Additional information

|                                                                                     |                                                                          |
|-------------------------------------------------------------------------------------|--------------------------------------------------------------------------|
| Proper shipping name                                                                | CHLOROANILINES, SOLID                                                    |
| Particulars in the transport document                                               | UN2018, CHLOROANILINES, SOLID, 6.1, II, (D/E), environmentally hazardous |
| Classification code                                                                 | T2                                                                       |
| Danger label(s)                                                                     | 6.1, "Fish and tree"                                                     |
|  |                                                                          |
| Environmental hazards                                                               | yes (hazardous to the aquatic environment)                               |
| Special provisions (SP)                                                             | 802(ADN)                                                                 |
| Excepted quantities (EQ)                                                            | E4                                                                       |
| Limited quantities (LQ)                                                             | 500 g                                                                    |

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|                               |     |
|-------------------------------|-----|
| Transport category (TC)       | 2   |
| Tunnel restriction code (TRC) | D/E |
| Hazard identification No      | 60  |
| <b>Emergency Action Code</b>  | 2X  |

### Regulations concerning the International Carriage of Dangerous Goods by Rail (RID)Additional information

|                            |                      |
|----------------------------|----------------------|
| <b>Classification code</b> | T2                   |
| <b>Danger label(s)</b>     | 6.1, "Fish and tree" |



|                              |                           |
|------------------------------|---------------------------|
| <b>Environmental hazards</b> | Yes<br>Hazardous to water |
|------------------------------|---------------------------|

|                                |          |
|--------------------------------|----------|
| <b>Special provisions (SP)</b> | 802(ADN) |
|--------------------------------|----------|

|                                 |    |
|---------------------------------|----|
| <b>Excepted quantities (EQ)</b> | E4 |
|---------------------------------|----|

|                                |       |
|--------------------------------|-------|
| <b>Limited quantities (LQ)</b> | 500 g |
|--------------------------------|-------|

|                                |   |
|--------------------------------|---|
| <b>Transport category (TC)</b> | 2 |
|--------------------------------|---|

|                                 |    |
|---------------------------------|----|
| <b>Hazard identification No</b> | 60 |
|---------------------------------|----|

### International Maritime Dangerous Goods Code (IMDG) - Additional information

|                                          |                                                          |
|------------------------------------------|----------------------------------------------------------|
| Proper shipping name                     | CHLOROANILINES, SOLID                                    |
| Particulars in the shipper's declaration | UN2018, CHLOROANILINES, SOLID, 6.1, II, MARINE POLLUTANT |
| Marine pollutant                         | yes (hazardous to the aquatic environment)               |
| <b>Danger label(s)</b>                   | 6.1, "Fish and tree"                                     |



|                          |          |
|--------------------------|----------|
| Special provisions (SP)  | -        |
| Excepted quantities (EQ) | E4       |
| Limited quantities (LQ)  | 500 g    |
| EmS                      | F-A, S-A |
| Stowage category         | A        |

### International Civil Aviation Organization (ICAO-IATA/DGR) - Additional information

|                                          |                                            |
|------------------------------------------|--------------------------------------------|
| Proper shipping name                     | Chloroanilines, solid                      |
| Particulars in the shipper's declaration | UN2018, Chloroanilines, solid, 6.1, II     |
| Environmental hazards                    | yes (hazardous to the aquatic environment) |
| <b>Danger label(s)</b>                   | 6.1                                        |



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Excepted quantities (EQ)

E4

Limited quantities (LQ)

1 kg

## SECTION 15: Regulatory information

### 15.1 Safety, health and environmental regulations/legislation specific for the substance or mixture

#### Relevant provisions of the European Union (EU)

##### Seveso Directive

| 2012/18/EU (Seveso III) |                                       |                                                                                       |       |
|-------------------------|---------------------------------------|---------------------------------------------------------------------------------------|-------|
| No                      | Dangerous substance/hazard categories | Qualifying quantity (tonnes) for the application of lower and upper-tier requirements | Notes |
| H2                      | acute toxic (cat. 2 + cat. 3, inhal.) | 50 200                                                                                | 41)   |

##### Notation

- 41) - Category 2, all exposure routes  
- category 3, inhalation exposure route

##### Deco-Paint Directive

|             |           |
|-------------|-----------|
| VOC content | 100 %     |
| VOC content | 1.430 g/l |

##### Industrial Emissions Directive (IED)

|             |       |
|-------------|-------|
| VOC content | 0 %   |
| VOC content | 0 g/l |

##### Directive on the restriction of the use of certain hazardous substances in electrical and electronic equipment (RoHS)

not listed

##### Regulation concerning the establishment of a European Pollutant Release and Transfer Register (PRTR)

not listed

##### Water Framework Directive (WFD)

| List of pollutants (WFD) |                                                                                                                                                                                                                                                                                   |        |           |         |
|--------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|-----------|---------|
| Name of substance        | Name acc. to inventory                                                                                                                                                                                                                                                            | CAS No | Listed in | Remarks |
| 4-Chloroaniline          | Organohalogen compounds and substances which may form such compounds in the aquatic environment                                                                                                                                                                                   |        | a)        |         |
| 4-Chloroaniline          | Substances and preparations, or the breakdown products of such, which have been proved to possess carcinogenic or mutagenic properties or properties which may affect steroidogenic, thyroid, reproduction or other endocrine-related functions in or via the aquatic environment |        | a)        |         |

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### Legend

a) Indicative list of the main pollutants

### Regulation on the marketing and use of explosives precursors

not listed

### Regulation on drug precursors

not listed

### Regulation on substances that deplete the ozone layer (ODS)

not listed

### Regulation concerning the export and import of hazardous chemicals (PIC)

not listed

### Regulation on persistent organic pollutants (POP)

not listed

### National regulations(GB)

### List of substances subject to authorisation (GB REACH, Annex 14) / SVHC - candidate list

not listed

### Restrictions according to GB REACH, Annex 17

| Dangerous substances with restrictions (GB REACH, Annex 17) |                        |        |    |
|-------------------------------------------------------------|------------------------|--------|----|
| Name of substance                                           | Name acc. to inventory | CAS No | No |
| 4-Chloroaniline                                             | carcinogenic           |        | 28 |

### Other information

Directive 94/33/EC on the protection of young people at work. Observe employment restrictions under the Maternity Protection Directive (92/85/EEC) for expectant or nursing mothers.

### National inventories

| Country | Inventory  | Status                       |
|---------|------------|------------------------------|
| AU      | AIIC       | substance is listed          |
| CA      | DSL        | substance is listed          |
| CN      | IECSC      | substance is listed          |
| EU      | ECSI       | substance is listed          |
| EU      | REACH Reg. | substance is listed          |
| JP      | CSCL-ENCS  | substance is listed          |
| KR      | KECI       | substance is listed          |
| MX      | INSQ       | substance is listed          |
| NZ      | NZIoC      | substance is listed          |
| PH      | PICCS      | substance is listed          |
| TW      | TCSI       | substance is listed          |
| US      | TSCA       | substance is listed (ACTIVE) |

### Legend

AIIC Australian Inventory of Industrial Chemicals  
CSCL-ENCS List of Existing and New Chemical Substances (CSCL-ENCS)  
DSL Domestic Substances List (DSL)  
ECSI EC Substance Inventory (EINECS, ELINCS, NLP)  
IECSC Inventory of Existing Chemical Substances Produced or Imported in China  
INSQ National Inventory of Chemical Substances  
KECI Korea Existing Chemicals Inventory  
NZIoC New Zealand Inventory of Chemicals

# Safety data sheet Safety data sheet

acc. to Regulation (EC) No. 1907/2006 (REACH)



## 4-Chloroaniline ≥98 %, for synthesis

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### Legend

PICCS Philippine Inventory of Chemicals and Chemical Substances (PICCS)  
REACH Reg. REACH registered substances  
TCSI Taiwan Chemical Substance Inventory  
TSCA Toxic Substance Control Act

## 15.2 Chemical safety assessment

No Chemical Safety Assessment has been carried out for this substance.

## SECTION 16: Other information

### Indication of changes (revised safety data sheet)

| Section | Former entry (text/value) | Actual entry (text/value)                              | Safety-relevant |
|---------|---------------------------|--------------------------------------------------------|-----------------|
| 15.1    |                           | National inventories:<br>change in the listing (table) | yes             |

### Abbreviations and acronyms

| Abbr.     | Descriptions of used abbreviations                                                                                                                                                    |
|-----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ADR       | Accord relatif au transport international des marchandises dangereuses par route (Agreement concerning the International Carriage of Dangerous Goods by Road)                         |
| ATE       | Acute Toxicity Estimate                                                                                                                                                               |
| CAS       | Chemical Abstracts Service (service that maintains the most comprehensive list of chemical substances)                                                                                |
| Ceiling-C | Ceiling value                                                                                                                                                                         |
| DGR       | Dangerous Goods Regulations (see IATA/DGR)                                                                                                                                            |
| EC No     | The EC Inventory (EINECS, ELINCS and the NLP-list) is the source for the seven-digit EC number, an identifier of substances commercially available within the EU (European Union)     |
| ED        | Endocrine disruptor                                                                                                                                                                   |
| EH40/2005 | EH40/2005 Workplace exposure limits ( <a href="http://www.nationalarchives.gov.uk/doc/open-government-licence/">http://www.nationalarchives.gov.uk/doc/open-government-licence/</a> ) |
| EINECS    | European Inventory of Existing Commercial Chemical Substances                                                                                                                         |
| ELINCS    | European List of Notified Chemical Substances                                                                                                                                         |
| EmS       | Emergency Schedule                                                                                                                                                                    |
| GB CLP    | The Chemicals (Health and Safety) and Genetically Modified Organisms (Contained Use) (Amendment etc.) (EU Exit) Regulations 2019, SI 2019/720 (as amended)                            |
| GB REACH  | The REACH etc. (Amendment etc.) (EU Exit) Regulations 2019, SI 2019/758 (as amended)                                                                                                  |
| GHS       | "Globally Harmonized System of Classification and Labelling of Chemicals" developed by the United Nations                                                                             |
| IATA      | International Air Transport Association                                                                                                                                               |
| IATA/DGR  | Dangerous Goods Regulations (DGR) for the air transport (IATA)                                                                                                                        |
| ICAO      | International Civil Aviation Organization                                                                                                                                             |
| ICAO-TI   | Technical instructions for the safe transport of dangerous goods by air                                                                                                               |
| IMDG      | International Maritime Dangerous Goods Code                                                                                                                                           |
| IMDG-Code | International Maritime Dangerous Goods Code                                                                                                                                           |
| index No  | The Index number is the identification code given to the substance in Part 3 of Annex VI to Regulation (EC) No 1272/2008                                                              |
| NLP       | No-Longer Polymer                                                                                                                                                                     |
| PBT       | Persistent, Bioaccumulative and Toxic                                                                                                                                                 |

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| Abbr. | Descriptions of used abbreviations                                                                                                                                      |
|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| REACH | Registration, Evaluation, Authorisation and Restriction of Chemicals                                                                                                    |
| RID   | Règlement concernant le transport International ferroviaire des marchandises Dangereuses (Regulations concerning the International carriage of Dangerous goods by Rail) |
| STEL  | Short-term exposure limit                                                                                                                                               |
| TWA   | Time-weighted average                                                                                                                                                   |
| VOC   | Volatile Organic Compounds                                                                                                                                              |
| vPvB  | Very Persistent and very Bioaccumulative                                                                                                                                |
| WEL   | Workplace exposure limit                                                                                                                                                |

### Key literature references and sources for data

Agreement concerning the International Carriage of Dangerous Goods by Road (ADR). Regulations concerning the International Carriage of Dangerous Goods by Rail (RID). International Maritime Dangerous Goods Code (IMDG). Dangerous Goods Regulations (DGR) for the air transport (IATA).

### List of relevant phrases (code and full text as stated in section 2 and 3)

| Code | Text                                                  |
|------|-------------------------------------------------------|
| H301 | Toxic if swallowed.                                   |
| H311 | Toxic in contact with skin.                           |
| H317 | May cause an allergic skin reaction.                  |
| H331 | Toxic if inhaled.                                     |
| H350 | May cause cancer.                                     |
| H400 | Very toxic to aquatic life.                           |
| H410 | Very toxic to aquatic life with long lasting effects. |

### Disclaimer

This information is based upon the present state of our knowledge. This SDS has been compiled and is solely intended for this product.